Industrial Predictive & Prescriptive Maintenance Intelligence Platform

Engine Health & Maintenance Decision-Support Report

Generated	2026-11-02 12:05
Dataset	Official test set (FD001)
Engine identifier	34
Last observed cycle	203
Input mode	Compatible engine trajectory
Model	XGBoost (version 20261101-231439)

5.9	**CRITICAL**
estimated RUL (cycles)	derived state (Healthy > 60; Warning 30-60; Critical < 30)

1. Executive summary

At cycle 203, the model estimates a remaining useful life of 5.9 cycles for engine 34, which maps to the derived state **Critical** under the configured thresholds. The separate anomaly detector flagged 100% of the last 10 cycles. Highest-priority recommendation: **Escalate for engineering review and plan an inspection** (Critical). Official ground-truth RUL for this test engine is 7 cycles (raw); shown for evaluation only.

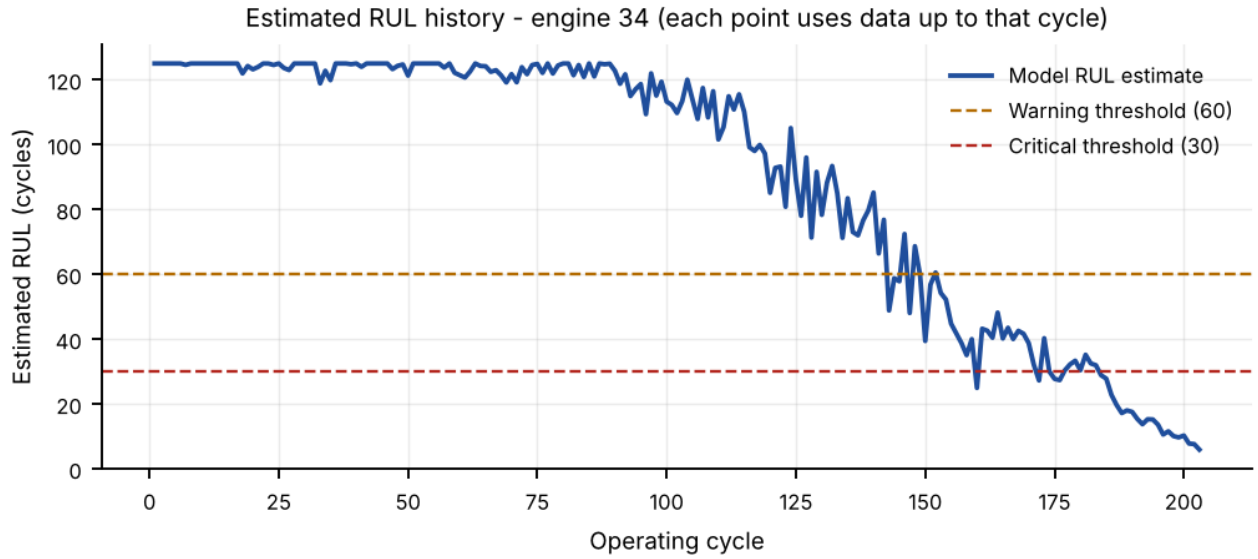
2. Data quality and input compatibility

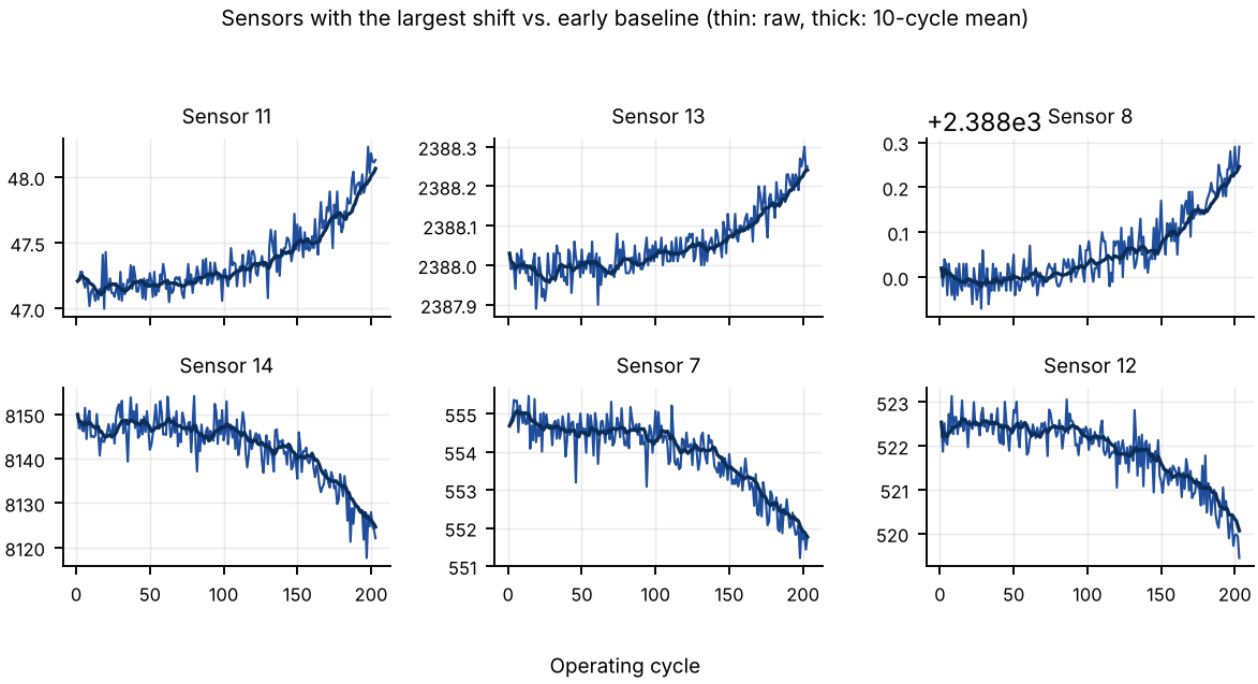
Validation status: **PASSED**. 203 cycles of history for this engine.

Severity	Message
Info	203 rows, 1 engine(s); cycles per engine 203–203.

Out-of-distribution check: 1 of 70 engineered features outside the training range (1%).

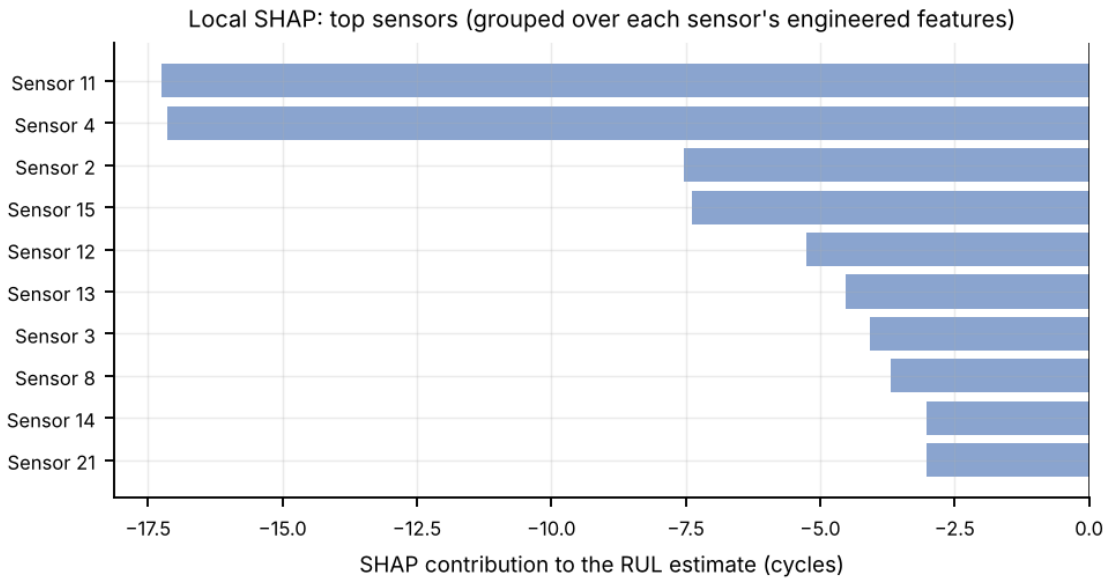
3. Estimated RUL history and sensor behaviour





Sensor	Baseline mean	Recent mean	Shift (σ of early noise)	Trend ρ
Sensor 11	47.24	48.06	+15.9	+0.85
Sensor 13	2388	2388	+15.4	+0.83
Sensor 8	2388	2388	+14.7	+0.84
Sensor 14	8148	8125	-14.3	-0.77
Sensor 7	555	551.8	-13.8	-0.84
Sensor 12	522.2	520.1	-12.5	-0.83
Sensor 15	8.376	8.495	+11.4	+0.78
Sensor 9	9069	9046	-10.3	-0.65

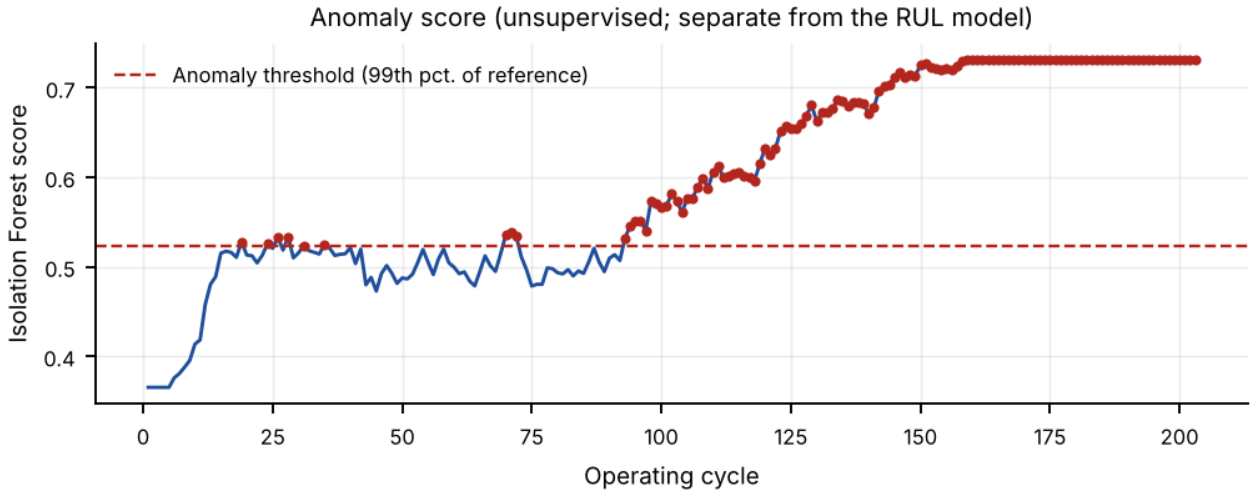
4. Explainability (SHAP)



Largest grouped contributions to this model's RUL estimate relative to the average training prediction (86.8 cycles): Sensor 11 (lowering the estimate by 17.3 cycles); Sensor 4 (lowering the estimate by 17.1 cycles); Sensor 2 (lowering the estimate by

7.5 cycles). These are statistical attributions of the model, not proof of a physical cause.
Explanation computed for the latest cycle's engineered features. SHAP describes model attribution, not physical causation.

5. Anomaly analysis (unsupervised, separate from the RUL model)



An anomaly means the sensor pattern is unusual relative to early-life reference behaviour. It is not automatically a failure; no fault labels exist to validate this detector.

6. Recommended inspection actions and supporting evidence

R-RUL-CRIT-E34 - Escalate for engineering review and plan an inspection

Recommended action	Escalate to the responsible maintenance engineer; review the engine's recent operating and sensor history and plan an engineering inspection according to the applicable maintenance procedure.
Triggering condition	Predicted RUL below the configured critical threshold.
Supporting evidence	Predicted RUL 5.9 cycles < critical threshold 30.
Finding type	Model-based finding + rule
Priority / evidence strength	Critical / Strong
Rule source	Application-configured rule (decision-support convention; not an industry standard)
Limitations	The threshold is a configurable convention. The model estimate has error (see Model Performance) and is capped; it does not identify which component is affected.
Suggested verification	Compare against the maintenance record and perform the inspection defined by the applicable procedure.
Action status	Pending

R-ANOM-PERSIST-E34 - Review unusual behaviour flagged by the anomaly detector

Recommended action	Review the engine's operating conditions and sensor integrity for the flagged period; potential contributing pattern identified — further engineering inspection is required to determine the physical cause.
Triggering condition	Anomaly flags in at least the configured fraction of recent cycles.
Supporting evidence	100% of the last 10 cycles exceed the anomaly threshold (latest score 0.731; first flagged at cycle 19).
Finding type	Rule-triggered (statistical/data-quality)
Priority / evidence strength	Medium / Moderate
Rule source	Application-configured rule (decision-support convention; not an industry standard); unsupervised Isolation Forest (see README)
Limitations	An anomaly is not automatically a failure: it can reflect degradation, an unusual operating condition or a measurement issue. The detector has no fault labels.
Suggested verification	Check sensor calibration/records and compare with peer engines over the same cycles.
Action status	Pending

R-SENSOR-SHIFT-E34 - Investigate sustained sensor drift

Recommended action	Potential contributing pattern identified in the listed sensors. Review the trends with an engineer who knows the sensor configuration; further inspection is required to determine the physical cause.
Triggering condition	One or more sensors shifted by more than the sigma trigger relative to the engine baseline.
Supporting evidence	Sensors shifted vs. the engine's early baseline by more than 3σ of early-life noise: Sensor 11: +15.9σ shift (trend ρ=+0.85); Sensor 13: +15.4σ shift (trend ρ=+0.83); Sensor 8: +14.7σ shift (trend ρ=+0.84); Sensor 14: -14.3σ shift (trend ρ=-0.77).
Finding type	Rule-triggered (statistical/data-quality)
Priority / evidence strength	Medium / Moderate
Rule source	Dataset documentation: Saxena et al., 'Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation', PHM08 (simulated data; no component mapping for sensors)
Limitations	Sensors in C-MAPSS are not mapped to physical components in the supplied documentation; a shift does not identify a faulty part.
Suggested verification	Validate the sensors against engineering documentation and maintenance records.
Action status	Pending

7. Model and evaluation context (population metrics, not this engine)

Model	XGBoost
Feature set	D: C + shift vs engine baseline
RUL cap (training/eval convention)	125 cycles
Validation	Engine-grouped 5-fold CV on the training file; official test set evaluated once at final cycles.
Internal CV (all cycles)	MAE 8.71, RMSE 12.58 cycles, R^2 0.909
Official test (final cycles, capped truth)	MAE 9.58, RMSE 13.66 cycles, R^2 0.884, NASA score 275 (n=100)

8. Limitations and required engineering verification

- The data are simulated (NASA C-MAPSS FD001); results do not demonstrate performance on real equipment.
- RUL estimates are model outputs with measurable error and are capped; they are not failure dates.
- Derived states use configurable thresholds and are not NASA labels or industrial standards.
- Sensors are not mapped to physical components; no physical root cause is identified or implied.
- All recommendations require review and confirmation by an authorised maintenance engineer.

Disclaimer. This application is a decision-support prototype built on the simulated NASA C-MAPSS dataset. It supports, and does not replace, the judgement of an authorised maintenance engineer or any certified industrial safety system. Predictions are statistical model estimates; explanations describe model attribution, not physical causation.